

Networking Assignment – 03

1. An ISP has been assigned the following four contiguous networks:

172.16.4.0/24, 172.16.5.0/24, 172.16.6.0/24, and 172.16.7.0/24.

The network engineer wants to advertise them as a single aggregated route. Determine the **supernet address with CIDR notation** and explain how route aggregation reduces the size of the routing table.

2. A router has the following four contiguous routing table entries:

10.20.8.0/24, 10.20.9.0/24, 10.20.10.0/24, and 10.20.11.0/24.

If these networks are summarized into a single route, determine the **new CIDR notation and subnet mask**.

3. A core router receives a packet with destination IP address **172.16.25.70**. Its forwarding table contains the following overlapping routes:

- 172.16.0.0/16 → Interface 1
- 172.16.16.0/20 → Interface 2
- 172.16.24.0/21 → Interface 3
- 172.16.25.64/26 → Interface 4

Determine **which interface the router will use** to forward the packet and explain your answer using the **Longest Prefix Match** rule.

4. A router receives a packet destined for **192.168.50.130**. The forwarding table contains:

- 192.168.0.0/16 → Interface A
- 192.168.48.0/20 → Interface B
- 192.168.50.0/24 → Interface C
- 192.168.50.128/27 → Interface D

Determine **all matching routes** and identify the **final forwarding interface** according to the Longest Prefix Match rule.

5. An organization owns the following eight contiguous networks:

**172.20.16.0/24, 172.20.17.0/24, 172.20.18.0/24, 172.20.19.0/24,
172.20.20.0/24, 172.20.21.0/24, 172.20.22.0/24, and 172.20.23.0/24.**

The network administrator wants to advertise all eight networks using **one supernet route**. Determine:

- (a) Supernet address
- (b) CIDR notation
- (c) Subnet mask
- (d) Total number of IP addresses represented by the supernet
- (e) Explain how this aggregation improves routing efficiency.

6. Link State Routing –

Consider the following network:

- A–B = 4
- A–C = 2

- B–C = 1
- B–D = 5
- C–D = 8
- C–E = 10
- D–E = 2

Using **Link State Routing and Dijkstra's Shortest Path First (SPF) algorithm**, determine the **shortest path from Router A to Router E**. Show the intermediate steps of Dijkstra's algorithm and construct the final shortest-path information.

7. Distance Vector vs. Link State Routing

Prepare a comparative analysis of **Distance Vector Routing** and **Link State Routing**. Your answer should include:

- Basic working principle
- Information exchanged between routers
- Routing algorithm used
- Convergence speed
- Routing table/information maintained
- Bandwidth and CPU requirements
- Routing problem
- Major advantages and disadvantages